

SAARTHAK KAPSE

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Education

Stony Brook University, NY, USA

Feb, 2021 – Mar, 2026

Doctor of Philosophy, Biomedical Informatics

Topic: Pretraining and Interpretability Techniques for Computational Pathology [\[Link\]](#)

Advised by: Prateek Prasanna, Dimitris Samaras, Joel Saltz

Collaborators: Srijan Das (Assistant Professor at UNC Charlotte, USA), Pushpak Pati (Senior Data Scientist at JNJ, Switzerland), Rajarsi Gupta

(Assistant Professor at Stony Brook University, USA), Chao Chen (Associate Professor at Stony Brook University, USA), Maria Vakalopoulou (Assistant

Professor at CentraleSupélec, University Paris Saclay, France)

Indian Institute of Technology, Bombay, India

Aug, 2016 – Dec, 2020

Bachelor of Technology, Electrical Engineering

Publications (selected)

- GenBio-PathFM: A State-of-the-Art Foundation Model for Histopathology** [\[Link\]](#) *under review*
Saarthak Kapse*, Mehmet Aygün*, Elijah Cole*, Emma Lundberg, Le Song, Eric P. Xing
- TICON: A Slide-Level Tile Contextualizer for Histopathology Representation Learning** [\[Link\]](#) *CVPR 2026 (findings)*
Saarthak Kapse*, V. Belagali*, P. Marza, S. Das, Z. Li, S. Boutaj, P. Pati, S. Yellapragada, T. Nandi, R. Madduri, J. Saltz, P. Prasanna, S. Christodoulidis, M. Vakalopoulou, D. Samaras
- Fast Vision Mamba: Pooling Spatial Dimensions for Accelerated Processing** [\[Link\]](#) *WACV 2026 (oral)*
Saarthak Kapse, R. Betz, S. Sivanandan
- Gigapixel Vision-Concept Contrastive Pretraining in Histopathology** [\[Link\]](#) *ICCV 2025 (highlight)*
Saarthak Kapse*, P. Pati*, S. Yellapragada, S. Das, R. Gupta, J. Saltz, D. Samaras, P. Prasanna
- SI-MIL: Taming Deep MIL for Self-Interpretability in Gigapixel Histopathology** [\[Link\]](#) *CVPR 2024*
Saarthak Kapse*, P. Pati*, S. Das, J. Zhang, C. Chen, M. Vakalopoulou, J. Saltz, D. Samaras, R. Gupta, P. Prasanna
- Attention de-sparsification matters: Inducing diversity in pathology representation learning** [\[Link\]](#) *MedIA 2023*
Saarthak Kapse, S. Das, J. Zhang, R. Gupta, J. Saltz, D. Samaras, P. Prasanna
- Prompt-MIL: Boosting Multi-Instance Learning via Task-specific Prompt Tuning** [\[Link\]](#) *MICCAI 2023*
J. Zhang, Saarthak Kapse, K. Ma, P. Prasanna, J. Saltz, M. Vakalopoulou, D. Samaras

Experience

Research Scientist | GenBio AI, Palo Alto, USA

Apr, 2026 – Present

- Developing multimodal foundation models that jointly learn from imaging, transcriptomics, and multiple other modalities
- Designing cross-modal pretraining and alignment objectives for paired and unpaired biological data

Graduate Researcher | IMAGINE Lab, Stony Brook University, NY, USA

Feb, 2021 – Mar, 2026

- Slide-level pretraining methods: Vision-Concept pretraining and a Slide-level Tile Contextualizer (CVPR'26; ICCV'25)
- Self-Interpretability in gigapixel Histopathology Using Handcrafted Features Grounded Model (CVPR'24)
- Domain-Driven Self-Supervised Learning in Computational Pathology (MedIA'23; IPMI'23; ISBI'23)

Research Intern | GenBio AI, Palo Alto, USA

Jun, 2025 – Dec, 2025

Built robust, data-efficient, and state-of-the-art histopathology foundation model (GenBio-PathFM) advised by *Elijah Cole*, *Emma Lundberg*, and *Le Song*

Research Intern | Insitro, San Francisco, USA

May, 2024 – Nov, 2024

Accelerated Vision Mamba with application in single-cell imaging advised by *Srinivasan Sivanandan* and *Justin Lee*

Undergraduate Researcher | Indian Institute of Technology, Bombay, India

Jan, 2020 – Dec, 2020

Gene Mutation Prediction from non-small cell lung cancer histopathology slides advised by *Amit Sethi*

Research Intern | Philips Innovation Campus, Bangalore, India

May, 2019 – Jul, 2019

Liver Lesion Segmentation from CT Scans using 3D Volumetric Deep Learning Approach advised by *M.S Dinesh*

Technical Skills

Technologies: Computer Vision, Deep Learning, Self-Supervised learning, Vision-Language Models, Generative Models

Domains: Natural Imaging, Computational Pathology, Medical Imaging (MRI, CT, X-Ray)

Languages & Frameworks: Python, PyTorch, TensorFlow, Hugging Face, WandB, QuPath, L^AT_EX